CSCI 2406: Topic 02

RISC Principles and ARM Architecture

RISC Design Principles, AArch64 Architectural Overview, Registers and Programmer-Visible State, and Instruction Classes

Computer Organization & Architecture | AArch64 Reference

Topic 2 Outline

Topic Objective

Examine the design principles commonly associated with RISC architectures and show how those principles appear concretely in AArch64.

- **Section 1**: RISC Design Principles
- **Section 2**: AArch64 Architectural Overview
- **Section 3**: Registers and Programmer-Visible State
- **Section 4**: Core AArch64 Instruction Classes
- **Section 5**: Summary and Self-Test

Section 01

RISC Design Principles

What Does RISC Mean?

Reduced Instruction Set Computer

RISC is a design philosophy emphasizing **regularity**, **simple instruction forms**, and a clean separation between computation and memory access.

- RISC does **not** simply mean "a processor with very few instructions."
- The important idea is regular instruction structure.
- Arithmetic and logical operations are primarily register-based.
- Load and store instructions transfer data between registers and memory.

Foundational RISC Design Principles

Instruction Regularity

- Regular instruction formats.
- Consistent operand fields.
- Predictable instruction alignment.

Register-Centered Computation

- Arithmetic uses registers.
- Memory access uses loads and stores.
- Regularity simplifies decoding.

Implementation Benefit

These characteristics support regular instruction fetch, decode, and pipelined implementation.

The Load/Store Architectural Model

Separation of Memory Access and Computation

In a load/store architecture such as AArch64, arithmetic and logical instructions operate on registers. Memory is accessed using dedicated **load** and **store** instructions.

Data Processing

- Operands are registers.
- Results are written to registers.

```
add x1, x2, x3
```

Memory Access

- Loads read memory into registers.
- Stores write registers to memory.

```
ldr x1, [x0]
```

Load–Compute–Store in AArch64

Suppose the address in `x0` identifies a 64-bit value in memory that must be increased by 5.

```
ldr x1, [x0]                    // load memory value into x1
```

```
add x1, x1, #5            // compute using register operands
```

```
str x1, [x0]                  // store result back to memory
```

Architectural Rule

The `add` instruction does not directly read or modify a memory operand.

Visualizing Load/Store Separation

Takeaway

Loads and stores connect memory to registers; arithmetic instructions operate on register operands.

RISC and CISC: Different Design Traditions

Characteristic	CISC Tradition	RISC Tradition
Instruction format	Often variable length	Typically more regular
Memory operands	Some instructions combine memory access and computation	Load/store separation
Addressing	Often many addressing forms	Comparatively regular forms
Decode	Can require more complex parsing	Regular fields simplify decode
Example	x86-64	AArch64, RISC-V

RISC and CISC are architectural traditions, not simple measures of modern processor complexity.

Same Operation, Different ISA Style

Suppose a 64-bit memory value must be increased by 5.

x86-64 Style

Some x86-64 instructions permit memory operands directly:

```
add qword ptr [rax], 5
```

One instruction describes both memory access and arithmetic.

AArch64 Load/Store Style

```
ldr x1, [x0]
add x1, x1, #5
str x1, [x0]
```

Memory transfer and arithmetic are represented by separate instructions.

Section 02

AArch64 Architectural Overview

ARM and AArch64

ARM Architecture

ARM defines a family of instruction set architectures used in embedded systems, mobile computing, desktop systems, and servers.

- **AArch64** is the 64-bit execution state introduced with ARMv8-A.
- It provides a modern 64-bit programmer-visible register model.
- It follows a load/store design with regular instruction formats.
- Different processors can implement AArch64 using different microarchitectures.

AArch64 and A64 Terminology

AArch64

- The architecture / execution state.
- Defines programmer-visible state.
- Includes registers, memory behavior, and instruction semantics.

A64

- The instruction set encoding.
- Used while executing in AArch64 state.
- A64 instructions are fixed 32-bit words.

Course Terminology

Use **AArch64** by default. Use **A64** only when referring specifically to the instruction set encoding.

Key Characteristics of AArch64

Instruction Model

- Load/store architecture.
- A64 instructions are 32 bits wide.
- Instructions are 4-byte aligned.
- Register fields are regular.

Register Model

- 31 general-purpose registers.
- 64-bit `x` register views.
- 32-bit `w` register views.
- Special roles for `sp`, `xzr`, PC state, and NZCV.

A64 Instruction Format Characteristics

Fixed-Width Encoding

Every instruction in the A64 instruction set occupies exactly **32 bits**.

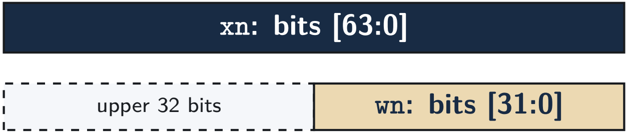
Operation	Operands	Options	Destination

- Fixed width simplifies instruction alignment and initial decode.
- Exact bit-field layouts vary by instruction class.

Detailed machine-code field layouts are covered later in the instruction-encoding topic.

Dual Register Views: `X` and `W`

Each general-purpose register can be accessed using either a 64-bit or 32-bit architectural view.



32-Bit Write Rule

Writing to a `wn` destination automatically clears bits [63:32] of the corresponding `xn` register.

Example: 32-Bit Write Clears Upper Bits

Instruction

```
add w3, w1, w2
```

- The operation uses the lower 32-bit views: `w1` and `w2`.
- The 32-bit result is written into `w3`.
- The upper half of `x3` is cleared automatically.

```
x3[63:32] <- 0
x3[31:0] <- result
```

Section 03

Registers and Programmer-Visible State

AArch64 General-Purpose Registers

Register Set

AArch64 provides **31 general-purpose registers**, named `x0` through `x30`. Each register is 64 bits wide.

64-Bit View

`x0`–`x30`

- Each `xn` holds 64 bits.
- Used for integer values, addresses, and general computation.

32-Bit View

`w0`–`w30`

- Each `wn` refers to bits [31:0] of `xn`.
- Writing `wn` clears bits [63:32].

Important

The ISA treats `x0`–`x30` as general-purpose registers. Software conventions assign additional roles during procedure calls.

The Zero Register

`xzr` / `wzr`

The zero register has two special behaviors: **reads return zero** and **writes are discarded**.

Use as a Zero Operand

```
mov x0, xzr
```

- Reading `xzr` produces the value `0`.
- Therefore, `x0` receives zero.

Use as a Discard Destination

```
cmp x1, x2
```

- `cmp` is an alias for `subs xzr, x1, x2`.
- The subtraction result is discarded, while NZCV flags are updated.

Key Idea

`xzr` behaves like a permanent zero value when read, and like a “trash destination” when written.

The Stack Pointer

sp

The stack pointer identifies the current stack location and has dedicated architectural semantics.

- Used by stack-related load/store and arithmetic forms.
- Represents a memory address, not a normal scratch register.
- Software conventions impose alignment and usage requirements.
- Stack behavior is studied more deeply with procedures and function calls.

```
stp x29, x30, [sp, #-16]!
```

Encoding Note: Register Number 31

Context-Dependent Meaning

In several A64 instruction formats, register encoding value 31 has special meaning.

Often Means Zero Register

`xzr` / `wzr`

Common in data-processing contexts.

Sometimes Means Stack Pointer

`sp` / `wsp`

Common in stack and memory-addressing contexts.

The instruction definition determines the meaning.

The Program Counter

Program Counter State

The program counter identifies the instruction address associated with the current flow of execution.

- Sequential execution advances to the next instruction.
- Branches and calls redirect execution to another address.
- Unlike AArch32 `r15`, the AArch64 program counter is **not** an ordinary general-purpose register.
- AArch64 arithmetic instructions cannot use `pc` as a normal register operand.

Condition Flags: NZCV

Some AArch64 instructions update four condition flags used by later operations.

N	Z	C	V
Negative	Zero	Carry	Overflow

- **N**: result has its sign bit set.
- **Z**: result is zero.
- **C**: carry-out for addition; no-borrow for subtraction.
- **V**: signed arithmetic overflow.

Programmer-Visible State: Summary

State	Architectural Role
`x0`–`x30`	31 general-purpose 64-bit registers
`w0`–`w30`	Lower 32-bit views of corresponding `x` registers
`xzr`/`wzr`	Zero operand; writes are discarded
`sp`	Stack pointer with dedicated semantics
Program Counter	Identifies control-flow position; not an ordinary GPR
NZCV	Condition flags used by flag-setting and conditional operations

Section 04

Core AArch64 Instruction Classes

Three Core Instruction Classes

For this course, core integer instructions are grouped into three broad functional classes.

Data Processing

- arithmetic
- logic
- comparison

`add`, `sub`
`and`, `cmp`

Memory Access

- loads
- stores
- transfer widths

`ldr`, `str`
`ldp`, `stp`

Control Flow

- branches
- calls
- returns

`b`, `b.eq`
`bl`, `ret`

Class 1: Data Processing

Data-processing instructions perform computation on register values.

Instruction	Operation	Effect
`add x3, x1, x2`	Addition	`x3 <- x1 + x2`
`sub x3, x1, x2`	Subtraction	`x3 <- x1 - x2`
`and x0, x1, x2`	Bitwise AND	`x0 <- x1 & x2`
`orr x0, x1, x2`	Bitwise OR	`x0 <- x1 | x2`

Key Characteristic

The operands are registers; these instructions do not directly read or write a memory operand.

Class 2: Memory Access

Memory instructions transfer data between registers and memory.

```
ldr x1, [x2]              // load 64 bits from memory into x1
```

```
str x5, [x3]              // store 64 bits from x5 to memory
```

```
ldr x1, [x2, #8]          // base address x2 plus byte offset 8
```

Bracket Notation

[x2] means: access memory using the address contained in x2.

Class 3: Control Flow

Control-flow instructions change which instruction executes next.

Unconditional

- `b target`: branch
- `bl function`: call
- `ret`: return

Conditional

- `b.eq target`: branch if equal
- `b.ne target`: branch if not equal
- Conditional branches may test NZCV flags.

Comparison and Branching

Common Pattern

```
cmp x0, x1
b.eq equal
```

- `cmp` compares two register values.
- It updates condition flags rather than storing a data result.
- `b.eq` uses those flags to decide whether to branch.

How the Instruction Classes Work Together

Consider:

$$*p = *p + 5$$

where `x0` contains the address represented by `p`.

Instruction	Class	Purpose
`ldr x1, [x0]`	Memory Access	Move the value from memory into a register
`add x1, x1, #5`	Data Processing	Compute the new value
`str x1, [x0]`	Memory Access	Write the result back to memory

RISC Connection

The sequence separates memory transfer from register computation.

Instruction Classification Exercise

Classify each instruction as **Data Processing**, **Memory Access**, or **Control Flow**.

Instruction	Your Classification
`add x3, x1, x2`	
`ldr x0, [x4]`	
`str w5, [x6]`	
`cmp x1, x2`	
`b.eq target`	
`bl function`	

Section 05

Summary & Self-Test

Topic 2 Summary

- **RISC Design**: regular formats, register-based computation, and explicit load/store memory access.
- **AArch64**: ARM's 64-bit architectural execution state.
- **A64**: fixed-width 32-bit instruction encoding used in AArch64 state.
- **Programmer-Visible State**: `x0`–`x30`, `sp`, `xzr`/`wzr`, PC state, and NZCV.
- **Instruction Classes**: Data Processing, Memory Access, and Control Flow.

Self-Test 1/3

1. What architectural characteristics are commonly associated with RISC designs?
2. What does it mean to say that AArch64 is a load/store architecture?
3. Why must a value normally be loaded into a register before an AArch64 arithmetic instruction can operate on it?
4. What is the distinction between **AArch64** and **A64**?

Self-Test 2/3

5. What happens to bits [63:32] of `x5` after an instruction writes to `w5`?
6. How general-purpose registers are visible in AArch64?
7. What are the roles of `xzr`, `sp`, and the program counter?
8. What do the `N`, `Z`, `C`, and `V` condition flags represent?

Self-Test 3/3

9. What are the three core instruction classes used in this course?
10. Classify each instruction: `add`, `ldr`, `str`, `cmp`, and `b.eq`.

Challenge

Explain how the sequence `ldr`, `add`, `str` demonstrates the RISC load/store model.

Wrap-Up and Next Steps

Topic 3: Machine-Level Data Representation

The next lecture examines how bit patterns manipulated by AArch64 instructions represent numerical values in registers and memory.

- Binary and hexadecimal representation
- Signed and unsigned integers
- Two's-complement representation
- Endianness and data alignment

Topic 2 established **what the AArch64 programmer can see**; Topic 3 examines **how data inside that visible state is represented**.